

Empirical Maximum Likelihood II

PHYS4730, F2025

Empirical Maximum Likelihood

- Assume (to simplify the math) we have an RV distribution of N integers.
- Centipede segments?
- Assuming conservation of probability and defining

$L :=$ “Empirical Likelihood”

we can derive the form of L_{MAX} .

$$\sum_{i=1}^N p_i = 1$$

$$L = \prod_{i=1}^N p_i$$

$$L_{MAX} = \prod_{i=1}^N \frac{1}{N}$$

Add a Constraint

- If the mean of the data $\langle x \rangle = \mu_0$, we need to add a **constraint** to the prior equations.
- At this point, our expression for L_{MAX} no longer works.
 - p_i in general *not* equal
 - Need to find new p_i , maximizing L , subject to the constraint.
- Recast L and the constraint equation with the use of a **Lagrange Multiplier λ** .

$$\sum_{i=1}^N p_i x_i = \mu_0 = \mu_0 \sum_{i=1}^N p_i$$

$$\sum_{i=1}^N p_i (x_i - \mu_0) = 0$$

$$\tilde{p}_i = \frac{1}{[1 + \lambda(x_i - \mu_0)]N}$$

$$\sum_{i=1}^N \frac{(x_i - \mu_0)}{[1 + \lambda(x_i - \mu_0)]} = 0$$

Lagrange Multipliers (Classical Mechanics)

Recall Lagrangian Mechanics. We define

$$L(y, \dot{y}, \theta, \dot{\theta}) = (\text{Kinetic Energy}) - (\text{Potential Energy})$$

then the calculus of variations tells us

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{y}} - \frac{\partial L}{\partial y} = 0$$

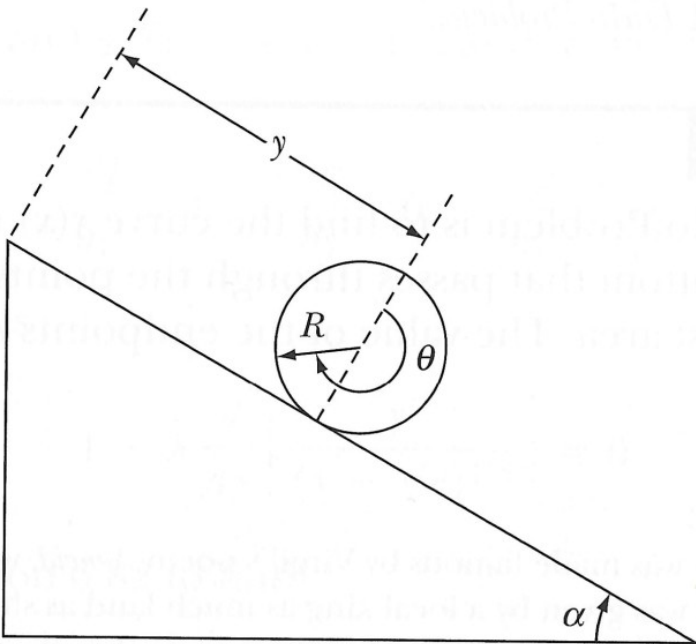
$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$$

But if the disc is rolling without slipping,
then we have an additional relationship
between the coordinates, namely

$$y = R\theta$$

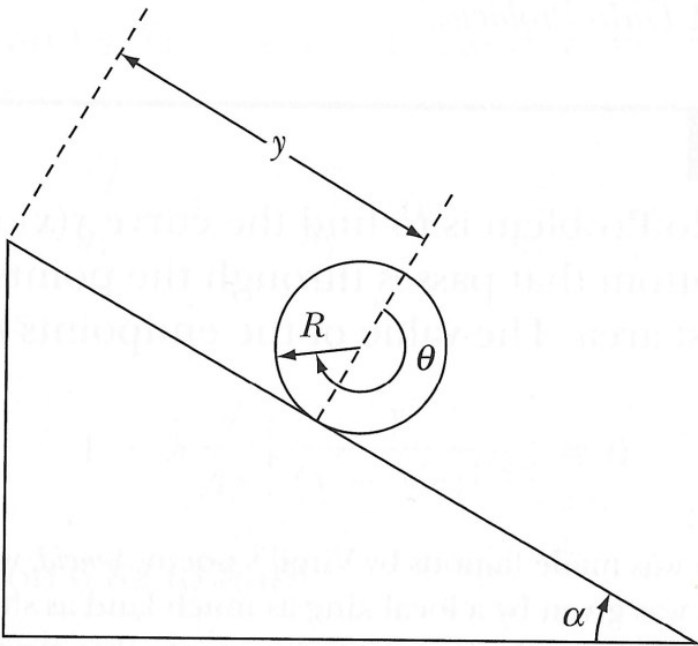
expressed as an *equation of constraint*

$$g(y, \theta) = y - R\theta = 0$$



Thornton & Marion

Lagrange Multipliers (Classical Mechanics)



Thornton & Marion

In such cases, it might be convenient to recast the equations of motion as

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{y}} - \frac{\partial L}{\partial y} = \lambda \frac{\partial g}{\partial y}$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = \lambda \frac{\partial g}{\partial \theta}$$

and the constraint equation

$$g(y, \theta) = y - R\theta = 0$$

Lagrange Multipliers (MELE)

Now, we can determine the mean μ_0 by maximizing L , or more typically

$$\frac{L}{L_{MAX}} = \frac{\prod_{i=1}^N \tilde{p}_i}{\prod_{i=1}^N \frac{1}{N}} = \prod_{i=1}^N N \tilde{p}_i$$
$$\ln \left(\frac{L}{L_{MAX}} \right) = - \sum_{i=1}^N \ln [1 + \lambda(x_i - \mu_0)]$$

A useful result from Owen is that the quantity

$$W = -2 \ln \left(\frac{L}{L_{MAX}} \right)$$
$$= 2 \sum_{i=1}^N \ln [1 + \lambda(x_i - \mu_0)]$$

will converge to a χ^2 distribution with $N_{df} = (\text{number of constraints})$. As shown in the `MELE_example.ipynb` notebook, this is useful in computing confidence intervals from the empirical likelihood ratio (ELR).

```
### function to evaluate empirical likelihood ratio ###  
  
def ELR(data,valu):  
  
    # Define the estimating function  
    g = data - valu  
  
    # Set up the Lagrange multiplier condition for Lam  
    constraintEQN = lambda L : sum(g/(1.0 + L*g))  
  
    # Find the multiplier  
    Lam = SciOpt.fsolve(constraintEQN,0)  
  
    # return the Empirical Likelihood Ratio W  
    W = 2.0*np.sum(np.log(1.0 + Lam*g))  
    return W  
  
#
```

Generalization of Empirical Maximum Likelihood

In our evaluation of the mean, we expressed the constraint as

$$\langle x \rangle = \mu_0$$

or more suggestively as

$$\langle x - \mu_0 \rangle = 0$$

We can regard

$$g(x, \mu_0) = x - \mu_0$$

as an “estimating function” for the mean, in that setting it’s expectation value to zero

$$\langle g \rangle = 0$$

results in an expression for determining the mean.

We can write estimating functions for other quantities as well. This is particularly easy for the higher moments, which are defined as averages of some quantity over the entire data set. For example, an estimating function for the variance (in the case of a known mean) would be

$$g(x, \mu, \sigma^2) = (x - \mu)^2 - \sigma^2$$

The procedure for extracting a quantity from the data using the ELR method is just to replace $(x - \mu_0)$ on the previous page with the relevant estimating function g , in particular

The constraint equation:

$$\sum_{i=1}^N \frac{g}{[1 + \lambda g]} = 0$$

The ELR:

$$W = 2 \sum_{i=1}^N \ln [1 + \lambda g]$$

In today’s lab you will practice this procedure by extracting the variance and it’s 95% confidence interval from the Weibull distribution.

Today’s Lab: